usepackage[a4paper,margin=0.65in]geometry

Collected Mathematical Papers of Arthur Cayley
Volume XIII — Pages 106–128
(Papers 915 (concl.), 916, 917, 918 (partial))

the summits to be numbered 1, 2, 3, ..., r in succession (the numbering may begin at any one of the r summits); regarding each of these numberings as giving a different partition, we should have the factor r. But, in fact, the partitions so obtained are not all of them distinct, but we have in each case a system of partitions repeated as many times as there are summits of the diagonals, that is, a number of times equal to twice the number of the diagonals; and we have thus, after the multiplication by r, to divide by the numbers 2, 4, 6, 8, in the four cases respectively.

21. We hence have immediately:—

Two parts, the number of partitions

$$= \frac{r}{2}A = \frac{r \,.\, r-3}{2 \,.\, 1};$$

Three parts, the number of partitions

$$= \frac{r}{2}A = \frac{r+1 \,.\, r \,.\, r-3 \,.\, r-4}{3 \,.\, 2 \,.\, 2 \,.\, 1};$$

Four parts, the number of partitions

$$= \frac{r}{6}(3A+2B) = \frac{r+2 \,.\, r+1 \,.\, r \,.\, r-3 \,.\, r-4 \,.\, r-5}{4 \,.\, 3 \,.\, 2 \,.\, 3 \,.\, 2 \,.\, 1},$$

the calculation being

$$\begin{aligned} 3\,(r^2+7r+2) &= 3r^2+21r+\ \ 6 \\ +\,2 \,.\, r-1 \,.\, r-2 &= +\,2r^2-\ \ 6r+\ \ 4 \\ \hline &\quad\ 5r^2+15r+10 \\ &= 5 \,.\, r+1 \,.\, r+2; \end{aligned}$$

Five parts, the number of partitions

$$= \frac{r}{8}(4A + 8B + 2C) = \frac{r+3\,.\,r+2\,.\,r+1\,.\,r\,.\,r-3\,.\,r-4\,.\,r-5\,.\,r-6}{5\,.\,4\,.\,3\,.\,2\,.\,4\,.\,3\,.\,2\,.\,1},$$

the calculation being

$$\begin{aligned}
4\,(r^3 + 18r^2 + 65r) &= 4r^3 + \;\;72r^2 + 260r \\
+\,8\,.\,r-1\,.\,r-2\,.\,r+7 &= +\,8r^3 + \;\;32r^2 - 152r + 112 \\
+\,2\,.\,r-1\,.\,r-2\,.\,r-7 &= +\,2r^3 - \;\;20r^2 + \;\;46r - \;\;28 \\
\hline
&\;\; 14r^3 + 84r^2 + 154r + \;\;84 \\
&= 14\,(r^3 + \;\;6r^2 + \;\;11r + \;\;6) \\
&= 14\,.\,r+1\,.\,r+2\,.\,r+3.
\end{aligned}$$

To complete the theory, it would be in the first instance necessary to find for any given number of diagonals, $k-1$, whatever, the number and form of the diagonal-types, A, B, C, &c.; this is itself an interesting question in the Theory of Partitions, but I have not considered it.

22. Although the foregoing process (which, it will be observed, deals with the diagonal-types, without any consideration of the sub-types) is the most simple for the determination of the numbers A, B, C, &c., yet it is interesting to give a second process. Considering the several cases in order:

One diagonal, A.—The diagonal has two summits; we must have on each side of it one summit, and there remain $r-4$ summits which may be distributed between the two intervals formed by the diagonals. This can be done in $\frac{r-3}{1}$ ways, or we have, as before,

$$A = \frac{r-3}{1}.$$

Two diagonals, A.—The diagonals have four summits; we must have outside each diagonal one summit, and there remain $r-6$ summits to be distributed between the four intervals formed by the diagonals; this can be done in $\frac{r-5\,.\,r-4\,.\,r-3}{6}$ ways, or we have this value for A°. But the two top summits of the diagonals, or the two bottom summits, may coalesce; in either case, the diagonals have three summits. We must have outside each diagonal one summit, and there remain $r-5$ summits to be distributed between the three intervals formed by the diagonals; the number of ways in which this can be done is

$$= \frac{r-4\,.\,r-3}{2},$$

say this is the value of A'. And we then have $A = A^\circ + 2A'$, as before. The calculation is

$$= \frac{r+1\,.\,r-3\,.\,r-4}{6}.$$

$$r-5+6 = r+1.$$

23. *Three diagonals, A.*—See No. 15 for the figures of the sub-types. We have

$$A = A^\circ + 4A' + 4A'',$$

where the coefficients, 4 and 4, are the number of ways in which A' and A'' respectively can be derived from A° by coalescences of summits. For A°, the diagonals have six summits, and there must be outside of two diagonals one summit; there remain $r-8$ summits to be distributed between the six intervals formed by the diagonals, and we have

$$A^\circ = \frac{r-7\,.\,r-6\,.\,r-5\,.\,r-4\,.\,r-3}{120}.$$

For A', the diagonals have five summits, and we must have outside of each of two diagonals, one summit; there remain $r-7$ summits to be distributed between the five intervals formed by the diagonals; we thus have

$$A' = \frac{r-6\,.\,r-5\,.\,r-4\,.\,r-3}{24}.$$

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For A'', the diagonals have four summits; there must be outside of each of two diagonals one summit, and there remain $r-6$ summits to be distributed between the four intervals formed by the diagonals; we thus have

$$A'' = \frac{r-5\,.\,r-4\,.\,r-3}{6}.$$

The foregoing values give

$$A = \frac{r-3\,.\,r-4\,.\,r-5}{120}\,(r^2+7r+2),$$

as before. The calculation is

$$\begin{array}{rrrr} r-6\,.\,r-7 = & r^2-13r & + & 42 \\ +20\,.\,r-6 = & +20r & - & 120 \\ +80 = & & + & 80 \\ \hline & r^2+\ \ 7r & + & 2. \end{array}$$

Three diagonals, B.—See No. 15 for the figures of the sub-types. We have

$$B = B^\circ + 3B' + 3B'' + B''',$$

For B°, the diagonals have six summits, and there must be outside each of the three diagonals one summit; there remain $r-9$ summits to be distributed between the six intervals formed by the diagonals. We thus have

$$B^\circ = \frac{r-8\,.\,r-7\,.\,r-6\,.\,r-5\,.\,r-4}{120}.$$

Similarly,

$$B' = \frac{r-7\,.\,r-6\,.\,r-5\,.\,r-4}{24},\quad B'' = \frac{r-6\,.\,r-5\,.\,r-4}{6},\quad B''' = \frac{r-5\,.\,r-4}{2}.$$

Hence

$$B = \frac{r-5\,.\,r-4\,.\,r-3\,.\,r-2\,.\,r-1}{120},$$

as before. The calculation is

$$\begin{array}{rl} r-6\,.\,r-7\,.\,r-8 = & r^3 - 21r^2 + 146r - \;\; 336 \\ +\,15\,.\,r-6\,.\,r-7 = & \qquad\quad + \; 15r^2 - 195r + \;\; 630 \\ +\,60\,.\,r-6 = & \qquad\qquad\quad + \; 60r - \;\; 360 \\ +\,60 = & \qquad\qquad\qquad\quad + \; 60 \\ \hline & r-1\,.\,r-2\,.\,r-3. \end{array}$$

24. *Four diagonals, A.*—The figures of the sub-types of A, B, C can be supplied without difficulty. We have

$$A = A^\circ + 6A' + 12A'' + 8A''',$$

where I remark that the numerical coefficients 1, 6, 12, 8 are the terms of $(1,\ 2)^3$. We have

$$A^\circ = \frac{r-9\,.\,r-8\,.\,r-7\,.\,r-6\,.\,r-5\,.\,r-4\,.\,r-3}{5040},$$

$$A' = \frac{r-8\,.\,r-7\,.\,r-6\,.\,r-5\,.\,r-4\,.\,r-3}{720},$$

$$A'' = \frac{r-7\,.\,r-6\,.\,r-5\,.\,r-4\,.\,r-3}{120},$$

$$A''' = \frac{r-6\,.\,r-5\,.\,r-4\,.\,r-3}{24},$$

and thence

$$A = \frac{r-3\,.\,r-4\,.\,r-5\,.\,r-6}{5040}\,(r^3+18r^2+65r),$$

as before. The calculation is

$$\begin{array}{rrrrr}
r-9\,.\,r-8\,.\,r-7 = & r^3 - & 24r^2 + & 191r - & 504 \\
+\,42\,.\,r-8\,.\,r-7 = & & +\ 42r^2 - & 630r + & 2352 \\
+\,504\,.\,r-7 = & & & +\ 504r - & 3528 \\
+\,1680 = & & & & +\ 1680 \\
\hline
& r^3 + & 18r^2 + & 65r. &
\end{array}$$

Four diagonals, B.—We have

$$B = B^\circ + 5B' + 9B'' + 7B''' + 2B^{\text{iv}},$$

where the coefficients, 1, 5, 9, 7, 2, are the terms of $(1,\ 1)^3(1,\ 2)$. We have

$$B^\circ = \frac{r-10\,.\,r-9\,.\,r-8\,.\,r-7\,.\,r-6\,.\,r-5\,.\,r-4}{5040},$$

$$B' = \frac{r-9\,.\,r-8\,.\,r-7\,.\,r-6\,.\,r-5\,.\,r-4}{720},$$

$$B'' = \frac{r-8\,.\,r-7\,.\,r-6\,.\,r-5\,.\,r-4}{120},$$

$$B''' = \frac{r-7\,.\,r-6\,.\,r-5\,.\,r-4}{24},$$

$$B^{\text{iv}} = \frac{r-6\,.\,r-5\,.\,r-4}{6},$$

and thence

$$B = \frac{r-3\,.\,r-4\,.\,r-5\,.\,r-6}{5040}\,\overline{r-1\,.\,r-2\,.\,r+7},$$

as before. The calculation is

$$\begin{array}{rl}
r-10\,.\,r-9\,.\,r-8\,.\,r-7 = & r^4 - \;\; 34r^3 + \;\; 431r^2 - \;\; 2414r + \;\;\; 5040 \\
+35\,.\,r-9\,.\,r-8\,.\,r-7 = & \qquad\quad + 35r^3 - \;\; 840r^2 + \;\; 6685r - 17640 \\
+378\,.\,r-8\,.\,r-7 = & \qquad\qquad\qquad + 378r^2 - \;\; 5670r + 21168 \\
+1470\,.\,r-7 = & \qquad\qquad\qquad\qquad\quad + 1470r - 10290 \\
+1680 = & \qquad\qquad\qquad\qquad\qquad\qquad + 1680 \\
\hline
& r^4 + \;\; r^3 - 31r^2 + 71r - \;\; 42 \\
& = r-3\,.\,r-2\,.\,r-1\,.\,r+7.
\end{array}$$

Four diagonals, C.—We have

$$C = C^\circ + 4C' + 6C'' + 4C''' + 1C^{\text{iv}},$$

where the coefficients are the terms of $(1,\,1)^4$. We have

$$C^\circ = \frac{r-11\,.\,r-10\,.\,r-9\,.\,r-8\,.\,r-7\,.\,r-6\,.\,r-5}{5040},$$

$$C' = \frac{r-10\,.\,r-9\,.\,r-8\,.\,r-7\,.\,r-6\,.\,r-5}{720},$$

$$C'' = \frac{r-9\,.\,r-8\,.\,r-7\,.\,r-6\,.\,r-5}{120},$$

$$C''' = \frac{r-8\,.\,r-7\,.\,r-6\,.\,r-5}{24},$$

$$C^{\text{iv}} = \frac{r-7\,.\,r-6\,.\,r-5}{6},$$

and thence

$$C = \frac{r-7\,.\,r-6\,.\,r-5\,.\,r-4\,.\,r-3\,.\,r-2\,.\,r-1}{5040}.$$

I omit the calculation, as the equation is at once seen to be a particular case of a known factorial formula.

25. We may analyse the partitions of an r-gon into a given number of parts, according to the nature of the parts, that is, the numbers of the sides of the several component polygons. It is for this purpose convenient to introduce the notion of "weight"; say a triangle has the weight 1, then a

quadrangle, as divisible into two triangles, has the weight 2, a pentagon, as divisible into three triangles, has the weight 3, $\ldots$, and generally an r-gon, as divisible into $r-2$ triangles, has the weight $r-2$. It at once follows that, if

$$W = w + w', \text{ or } = w + w' + w'', \text{ \&c.,}$$

then a polygon of weight W is divisible into two polygons of the weights w, w', or into three polygons of the weights w, w', w'' respectively; and so on. Thus the 2-partitions of an 8-gon (weight = 6) are 15, 24, and 33; the 3-partitions are 114, 123, 222, and so on. Of course the number of the partitions 15, 24, 33, is equal to the whole number of the 2-partitions of the 8-gon, that is, = 20; the number of the partitions 114, 123, 222, is equal to the whole number of the 3-partitions of the 8-gon, that is, it is = 120; and so in other cases. It is easy to derive in order one from the other the numbers of the partitions of each several kind of the polygons of the several weights 2, 3, 4, 5, 6, &c.; and I write down the accompanying Table (No. 26), facing page 112, the process for the construction being as follows:

27. The first column (2 parts) is at once obtained. For a polygon of an odd number of sides, for instance the 9-gon (weight = 7), imagining the summits numbered in order 1, 2, ..., 9, we divide this into a triangle and octagon, or obtain the partitions 16, by drawing the diagonals 13, 24, ..., 81, 92: viz. the number is = 9.

In the Table this is written, 16 = 9; and so in other cases. Similarly we divide it into a quadrangle and heptagon, or obtain the partitions 25, by drawing the diagonals 14, 25, ..., 82, 93: viz. the number is again = 9; and we divide it into a pentagon and a hexagon, or obtain the partitions 34, by drawing the diagonals 15, 26, ..., 83, 94: viz. the number is = 9, and here

$$9 + 9 + 9 = 27,$$

the whole number of 2-partitions of the 9-gon. For a polygon of an even number of sides, for instance the 10-gon (weight = 8), the process is a similar one, the only difference being that for the division into two hexagons, (that is, for the partitions 44), each partition is thus obtained twice, or the number of such partitions is $\frac{1}{2} \cdot 10$, = 5; the numbers for the partitions 17, 26, 35, 44, thus are 10, 10, 10, 5; and we have

$$10 + 10 + 10 + 5 = 35,$$

the whole number of the 2-partitions of the 10-gon.

28. To obtain the second column (3 parts)—suppose, for instance, the 3-partitions of the 9-gon; these are 115, 124, 133, 223. We obtain the number of the partitions 115 from the terms

$$16 = 9 \quad \text{and} \quad 25 = 9$$

of the first column: viz. in 16, changing the 6 into 15, that is, dividing the polygon of weight 6 into two parts of weights 1 and 5 respectively: this can be done in eight ways (see, higher up, $15 = 8$ in the first column); and we thus obtain the number of partitions

$$9 \times 8 = 72;$$

and again, in 25, changing the 2 into 11, that is, dividing the polygon of weight 2 into two parts each of weight 1: this can be done in two ways (see, higher up, $11 = 2$ in the first column); and we thus obtain the number of partitions

$$9 \times 2 = 18;$$

we should thus have, for the number of partitions 115, the sum

$$72 + 18 = 90,$$

only, as it is easy to see, each partition is obtained twice, and the number of the partitions 115 is the half of this, $= 45$. And by the like process it is found that the numbers of the partitions 124, 133, 223 are equal to 90, 45, 45 respectively; and then, as a verification, we have

$$45 + 90 + 45 + 45 = 225,$$

the whole number of the 3-partitions of the 9-gon.

29. The third column (4 parts) is derived in like manner from the second column by aid of the first column; and so in general, each column is derived in like manner from the column which immediately precedes it, by aid of the first column. And we have for the numbers in each compartment of any column the verification that the sum of these numbers is equal to the whole number (for the proper values of k and r) of the k-partitions of the r-gon.

It might be possible, by an application of the method of generating functions, to find a law for the numbers in any compartment of a column of the table; but I have not attempted to make this investigation.

30. In the table in No. 2, the numbers 1, 2, 5, 14, 42, &c., of the diagonal line show the number of partitions of the triangle, the quadrangle, the 5-gon, ..., r-gon into triangles: viz. these numbers show the number of partitions of the r-gon into $r-2$ parts, that is, into triangles; and, for the r-gon, writing $k = r - 2$, the number is

$$= \frac{[2r-4]^{r-3}}{[r-2]^{r-3}}.$$

If, as above, taking the weight of the triangle to be 1, we write

$$r - 2 = w,$$

then the number is

$$= \frac{[2w]^{w-1}}{[w]^{w-1}},$$

viz. this is the expression for the number of partitions of the polygon of weight w, or $(w + 2)$-gon, into triangles.

31. The question considered by Taylor and Rowe, in the paper referred to in No. 1, is that of the partition of the r-gon into p-gons, for p, a given number > 3; this implies a restriction on the form of r, viz. we must have $r-2$ divisible by $p-2$. In fact, generalizing the definition of w, if we attribute

to a p-gon the weight 1, and accordingly to a polygon divisible into w p-gons the weight w, then, r being the number of summits, we must have

$$r = (p - 2)\, w + 2.$$

In particular, if $p = 4$, so that the r-gon is to be divided into quadrangles, then r is necessarily even, and for the values

$$w = 1,\ 2,\ 3,\ \ldots,$$

we have

$$r = 4,\ 6,\ 8,\ \ldots.$$

[To face page 112.

r	W	7 PARTS	8 PARTS
3	1		
4	2		
5	3		
6	4		
7	5		
8	6		
9	7 57	1111111 $1287 \times 2 = 2574$ $6\,)\,\overline{2574} = 429$	
10	8 12 05 $\overline{07}$	1111112 $2002 \times 5 = 10010$ $5005 \times 4 = 20020$ $6\,)\,\overline{30030} = 5005$	11111111 $5005 \times 2 = 10010$ $7\,)\,\overline{10010} = 1430$

[Note: this is the surviving fragment of the original folding-table facing page 112; the remaining compartments are blank in the printed volume.]

32. To fix the ideas, I assume $p = 4$, and thus consider the problem of the division of the $(2w + 2)$-gon into quadrangles. Writing

$$w - 1 = a + b + c,$$

we take at pleasure any one side of the $(2w + 2)$-gon, making this the first side of a quadrangle, the second, third, and fourth sides being diagonals of the polygon such that outside the second side we have a polygon of weight a, outside the third side a polygon of weight b, and outside the fourth side a polygon of weight c. Any one or more of the numbers a, b, c may be $= 0$; they cannot be each of them $= 0$ except in the case $w = 1$. The meaning is that the corresponding side of the quadrangle, instead of being a diagonal, is a side of the $(2w + 2)$-gon, viz. there is no polygon outside such side. Suppose, in general, that P_w is the number of ways in which a polygon of weight w can be divided in quadrangles, and let each of the polygons of weights a, b, c respectively, be divided into quadrangles: the number of ways in which this can be done is $P_a P_b P_c$; and it is to be noticed that, if for instance $a = 0$, then the number is $= P_b P_c$, viz. the formula remains true if only we assume $P_0 = 1$. The number of partitions thus obtained is $\Sigma P_a P_b P_c$, where the summation extends to all the partitions of $w - 1$ into the parts a, b, c (zeros admissible and the order of the parts being attended to). And we thus obtain all the partitions of the $(2w + 2)$-gon into w parts; for first, the partitions so obtained are all distinct from each other, and next every partition of the $(2w + 2)$-gon into w parts is a partition in which the selected side of the $(2w + 2)$-gon is a side of some one of the quadrangles. That is, we have

$$P_w = \Sigma P_a P_b P_c \ (a,\ b,\ c \text{ as above});$$

and it hence appears that, considering the generating function

$$f = 1 + P_1 x + P_2 x^2 + P_3 x^3 + \ldots,$$

we have

$$f = 1 + x f^3.$$

The reasoning is precisely the same if, instead of a division into quadrangles, we have a division into p-gons; the only difference is that instead of the three parts a, b, c, we have the $p - 1$ parts a, b, c, $\ldots$, and the equation for f thus is

$$f = 1 + x f^{p-1}.$$

33. Writing for a moment

$$f = u + xf^{p-1},$$

and expanding by Lagrange's theorem, we have

$$f = u + \frac{x}{1}\,(u^{p-1}) + \frac{x^2}{1\,.\,2}\,(u^{2(p-1)})' + \ldots + \frac{x^w}{1\,.\,2\,\ldots\,w}\,(u^{w(p-1)})^{(w-1)} + \ldots,$$

viz. after the differentiation, writing $u = 1$, we have

$$P_w = \frac{[(p-1)\,w]^{w-1}}{[w]^{w-1}},$$

where it will be recollected that, for the number of sides of the polygon, we have

$$r = (p-2)\,w + 2.$$

In the case of the partition into triangles, $p = 3$, and we have the beforementioned value

$$[2w]^{w-1} \div [w]^{w-1}, \quad w = r - 2.$$

916.

[NOTE ON A THEOREM IN MATRICES.]

[From the *Proceedings of the London Mathematical Society*, vol. XXII. (1891), p. 458.]

PROF. CAYLEY remarks that a "simple instance [of the theorem] is that, if the real symmetric matrix

$$\begin{pmatrix} a, & h, & g \\ h, & b, & f \\ g, & f, & c \end{pmatrix}$$

has two latent roots each $= 0$, and therefore a vacuity $= 2$, then it has also a nullity $= 2$ [which may be shown as follows], viz. the conditions for a vacuity $= 2$ are

$$\begin{vmatrix} a, & h, & g \\ h, & b, & f \\ g, & f, & c \end{vmatrix} = 0, \quad bc - f^2 + ca - g^2 + ab - h^2 = 0,$$

or, if as usual the determinant is called K, and if

$$(A,\, B,\, C,\, F,\, G,\, H) = (bc - f^2,\, ac - g^2,\, \ldots),$$

then, if

$$K = 0, \quad A + B + C = 0,$$

these equations give

$$BC = F^2, \quad AC = G^2, \quad AB = H^2,$$

i.e.

$$BC = F^2, \quad AC = G^2, \quad AB = H^2;$$

and therefore

$$\begin{aligned} A\,(A+B+C) &= A^2+H^2+G^2, \\ B\,(A+B+C) &= H^2+B^2+F^2, \\ C\,(A+B+C) &= G^2+F^2+C^2; \end{aligned}$$

or, if $A+B+C=0$, then for real values

$$A=B=C=F=G=H=0,$$

i.e. nullity $=2$."

917.

[NOTE ON THE THEORY OF RATIONAL TRANSFORMATION.]

[From the *Proceedings of the London Mathematical Society*, vol. XXII. (1891), pp. 475, 476.]

In my paper, "Note on the Theory of the Rational Transformation between Two Planes, and on Special Systems of Points," *Proc. Lond. Math. Soc.* t. III. (1870), pp. 196–198, [450], I notice a difficulty which presents itself in the theory. The transformation is given by the equations

$$x' : y' : z' = X : Y : Z,$$

where X, Y, Z are functions $(\ast\!\!\!\rangle\, x,\ y,\ z)^n$, such that $X=0$, $Y=0$, $Z=0$ are curves in the first plane passing through α_1 points each once, α_2 points each twice (that is, having each of the α_2 points for a double point), α_3 points each 3 times, and so on. We have as the condition of a single variable point of intersection,

$$\alpha_1 + 4\alpha_2 + 9\alpha_3 + \ldots = n^2 - 1,$$

and as the condition in order that each of the curves $X=0$, $Y=0$, $Z=0$, or say the curve $aX + bY + cZ = 0$, may be unicursal,

$$\alpha_2 + 3\alpha_3 + \ldots = \tfrac{1}{2}(n-1)(n-2);$$

and we thence deduce

$$\alpha_1 + 3\alpha_2 + 6\alpha_3 + \ldots = \tfrac{1}{2}n\,(n+3) - 2;$$

viz. the postulation of the fixed points quoad a curve of the order n is less by 2 than the postulandum (or, as I prefer to call it, the capacity) $\frac{1}{2}n(n+3)$ of the curve of the order n; that is, there are precisely the three asyzygetic curves $X=0$, $Y=0$, $Z=0$. This is as it should be, assuming that the (α_1, α_2, α_3, …) points are an ordinary system of points: but what if they form a special system having a postulation less

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than $\alpha_1 + 3\alpha_2 + 6\alpha_3 + \ldots$? If, for instance, the postulation is $= \alpha_1 + 3\alpha_2 + 6\alpha_3 + \ldots - 1$, then this would be $= \frac{1}{2}n(n+3) - 3$, and there would be four asyzygetic curves $X = 0$, $Y = 0$, $Z = 0$, $W = 0$. I believe this to be impossible; but the only proof which I can offer rests upon a remark in regard to the form of the tables[1], pp. 148, 149, of my paper "On the Rational Transformation between Two Spaces," *Proc. Lond. Math. Soc.*, t. III. (1870), pp. 127–180, [447]. I recall that the Jacobian curve $J(X, Y, Z) = 0$ consists of α'_1 lines, α'_2 conics, α'_3 cubics, ..., &c., each passing a certain number of times through the $(\alpha_1, \alpha_2, \alpha_3, \ldots)$ points, and that the number of times of passage is shown by these tables; thus, *loc. cit.*, $n = 5$, $\alpha_1 = 8$, $\alpha_4 = 1$: the Jacobian consists of eight lines and a quartic, and we have the table $(\alpha'_1 = 8, \alpha'_4 = 1)$,

	α_1	α_2	α_3	α_4
	8	0	0	1
$\alpha'_1 = 8$	1			1
$\alpha'_2 = 0$				
$\alpha'_3 = 0$				
$\alpha'_4 = 1$	8			1^3

showing that the quartic passes through the eight points α_1, and through the point α_4 three times (has α_4 for a triple point). Imagine a new function W. Then in like manner $J(X, Y, W) = 0$ consists of eight lines and a quartic, and this quartic passes through the eight points α_1 and the point α_4 three times; that is, the two quartics intersect in $8 + 3 \cdot 3, = 17$ points; and thus the two quartics must be one and the same curve; this implies a syzygy between X, Y, Z, W, viz. W is a mere linear function of X, Y, Z. The general remark is that, if in the tables mp is reckoned as mp^2, then in the table for the several lines (exclusive of those for which the outside accented letter is $= 0$, and therefore the tabular numbers of the line are each $= 0$), i.e. for the lines which correspond to a line, a conic, a cubic, a quartic, &c., respectively, the sums of the tabular numbers are 1^2+1, 2^2+1, 3^2+1, 4^2+1, &c., respectively. This is, in fact, the case for each of the eleven tables (*loc. cit.*).

[1] [* This Collection, vol. VII., pp. 208, 209.]

918.

ON THE SUBSTITUTION GROUPS FOR TWO, THREE, FOUR, FIVE, SIX, SEVEN, AND EIGHT LETTERS.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. XXV. (1891), pp. 71–88, 137–155.]

THE substitution groups for two, three, four, and five letters were obtained by Serret: those for six, seven and eight letters have recently been obtained by Mr Askwith. I wish to reproduce these results in a condensed form.

The following table shows for the several cases respectively, the orders of the several groups, and for any order the number of distinct groups. As regards the case of eight letters, the numbers mentioned do not exactly agree with Mr Askwith: he gives a few non-existent groups, and omits some which I have supplied (see *post*, the list of the groups for eight letters); and it is possible that there are other omissions: the several numbers in the column and the sum total of 155 are given subject to correction.

2		3		4		5		6		7		8	
1	2	1	3	1	2	1	5	1	2	2	6	1	2
		1	6	3	4	2	6	1	3	1	7	8	4
				1	8	1	10	4	4	2	10	4	6
				1	12	1	12	3	6	9	12	25	8
				1	24	1	20	6	8	1	14	6	12
						1	60	1	9	2	20	1	15
						1	120	1	12	1	21	22	16
								1	16	7	24	3	18
								3	18	1	36	4	24
								4	24	1	40	3	30
								2	36	1	42	12	32
								2	48	1	48	5	36
								1	60	3	72	14	48
								1	72	2	120	3	60
								1	120	1	144	3	64
								1	360	1	240	3	72
								1	720	1	2520	10	96
										1	5040	3	120
												2	144
												1	168
												1	180
												3	192
												1	240
												3	288
												1	336
												3	360
												1	384
												2	576
												3	720
												1	1152
												1	1440
												1	20160
												1	40320
1		2		7		8		34		38		155	

Here the top line shows the numbers of letters; each second column shows the order of the groups, and each first column the number of groups of the several orders: the sums at the foot of the first columns show therefore the whole number of groups, viz.

No. of letters = 2, 3, 4, 5, 6, 7, 8

No. of groups = 1, 2, 7, 8, 34, 38, 155.

In the enumeration of the groups, I use some notations which must be explained. For greater simplicity I omit parentheses, and write ab, abc, $ab\,.\,cd$, $abc\,.\,def$, &c., to denote substitutions, viz. ab is the interchange of a and b; abc the cyclical change a into b, b into c, c into a; $ab\,.\,cd$ the combined interchange of a and b and of c and d; $abc\,.\,def$ the combined cyclical changes a into b, b into c, c into a, and d into e, e into f, f into d; and so in other cases.

Again, (abc) all, means the complete group of all the substitutions

$$(1,\ abc,\ acb,\ bc,\ ca,\ ab)$$

upon the three letters; and so in other cases. In the case, however, of two letters, I write simply (ab) to denote the complete group $(1,\, ab)$ of substitutions, and so also for any substitution such as $ab\,.\,cd$, where the complete group is $(1,\, ab\,.\,cd)$, I denote the group by $(ab\,.\,cd)$. Moreover, (abc) cyc., denotes the group of cyclical substitutions $(1,\, abc,\, acb)$ upon the three letters; and so in other cases.

A group will in general contain positive and negative substitutions, and, when this is so, the positive substitutions will form a group which is denoted by the symbol, POS.; the negative substitutions (which of course do not form a group) are denoted in like manner by the symbol, NEG. Thus, (abc) all, pos., or for shortness, (abc) pos., will denote the group formed by the positive substitutions of (abc) all; (abc) pos. is thus the same thing as (abc) cyc., but obviously $(abcd)$ pos. and $(abcd)$ cyc. have quite different meanings. It is to be noticed that, for any odd number of letters, the substitutions of a group (abc) cyc. are all positive, and thus (abc) cyc. pos. is the original group: for any even number of letters, the group $(abcd)$ cyc. or $(abcdef)$ cyc. contains positive and negative substitutions, but the positive substitutions thereof form a cyclical group, thus

$$(abcd)\,\text{cyc. pos.} = (ab\,.\,cd)\,\text{cyc.},\quad (abcdef)\,\text{cyc. pos.} = (ace\,.\,bdf)\,\text{cyc.},$$

and the notation, $(\quad)$ cyc. pos., is thus unnecessary, and it will not be used.

Substitutions or groups which have no letter in common are said to be independent. The product of two independent groups is of course a group, and the components may be called independent factors of the resultant group: we use for such a product the notation $A\,.\,B$, and call the group a composite group. If each of the groups A and B contain positive and negative substitutions, we thence derive a new group, $(A\,.\,B)$ pos., viz. the substitutions hereof are the products of a positive substitution of A and a positive substitution of B, and the products of a negative substitution of A and a negative substitution of B, say

$$(A\,.\,B)\,\text{pos.} = (A\,\text{pos.}\ B\,\text{pos.}) + (A\,\text{neg.})(B\,\text{neg.}) :$$

obviously the number of substitutions or order of the group $(A\,.\,B)$ pos. is one half of that of the group $A\,.\,B$.

A more general, but not perfectly definite, notation is that of $(A\,.\,B)$ dim., the dimidiate of the group $A\,.\,B$. Suppose, for instance, that A is a group of substitutions of the letters $(a,\, b,\, c,\, d)$; and that B is the group

(ef). Here if A is composed of two equal sets $1, P, Q, \ldots;\ R, S, T, \ldots$, where the first set $1, P, Q, \ldots$, is a group, then we have a group $1, P, Q, \ldots, ef.R, ef.S, ef.T, \ldots$, which is a group of the form in question, $(A.B)$ dim. But in some cases the group A can be in more than one way divided into two equal sets the first of which forms a group, and a further explanation of the notation is required. I do not give at present a more complete explanation, nor explain the analogous notions trisection (tris.), &c.

Substitutions or groups having a letter or letters in common are non-independent. If A, B are such groups, then the substitutions of A are not commutable with those of B, and we have not in general such a group as $B\,.\,A$ or $A\,.\,B$. It may happen that, although the individual substitutions of A are not commutable with those of B, yet that the groups are commutable, say we have $A\,.\,B = B\,.\,A$, viz. here the substitutions of $A\,.\,B$ are in a different order identical with those of $B\,.\,A$. We have in this case a group $A\,.\,B$; this is not a composite group; and the notation will never be employed without explanation.

I consider, in particular, the two groups $(abc\,.\,def)$ cyc. and (abc) cyc. (def) cyc.; in each of them, the six letters (a, b, c, d, e, f) are divided into two sets (a, b, c), (d, e, f), and the substitutions are the product of an abc-substitution into a def-substitution: viz. in the first group the substitutions are

$$1,\ abc\,.\,def,\ acb\,.\,dfe,$$

and in the second group they are

$$\begin{array}{rll} 1, & abc, & abc\,.\,def, \\ & acb, & acb\,.\,def, \\ def, & abc\,.\,dfe, & \\ dfe, & acb\,.\,dfe. & \end{array}$$

We can from each of the groups, introducing substitutions which interchange (a, b, c) with (d, e, f), or say by combination with the group $(ad\,.\,be\,.\,cf)$, derive a group of double the order; and it is worth while to consider the two cases in detail. First, for the group

$$(ad\,.\,be\,.\,cf)(abc\,.\,def)\text{ cyc.}$$

We have

Arrangements.	Substitutions.
$abcdef$,	1,
$bcaefd$,	$abc\,.\,def$,
$cabfde$,	$acb\,.\,dfe$,
$defabc$,	$ad\,.\,be\,.\,cf$,
$efdbca$,	$aecdbf$,
$fdecab$,	$afbdce$,

say

$$(ad\,.\,be\,.\,cf)(abc\,.\,def)\text{ cyc.} = (abcdef)_6.$$

This group of the order 6 is, as it happens, the group $(aecdbf)$ cyc.

It may be remarked that if, instead of $(abc\,.\,def)$ cyc., we consider the precisely similar group $(abc\,.\,dfe)$ cyc., then operating upon it with the same group $(ad\,.\,be\,.\,cf)$, we obtain quite a different form of group $(abcdef)_6$. In fact, for $(ad\,.\,be\,.\,cf)(abc\,.\,dfe)$ cyc., we have

Arrangements.	Substitutions.
$abcdef$,	1,
$bcafde$,	$abc \,.\, dfe$,
$cabefd$,	$acb \,.\, def$,
$defabc$,	$ad \,.\, be \,.\, cf$,
$efdcab$,	$ae \,.\, bf \,.\, cd$,
$fdebca$;	$af \,.\, bd \,.\, ce$;

which is not the cyclical group. See *post*, six letters, ord. 6. 3.

Next, for the group $(ad \,.\, be \,.\, cf)(abc)$ cyc. (def) cyc.; we have here

Arrangements.	Substitutions.
$abcdef$,	1,
$bcadef$,	abc,
$cabdef$,	acb,
$abcefd$,	def,
$bcaefd$,	$abc \,.\, def$,
$cabefd$,	$acb \,.\, def$,
$abcfde$,	dfe,
$bcafde$,	$abc \,.\, dfe$,
$cabfde$,	$acb \,.\, dfe$,
$defabc$,	$ad \,.\, be \,.\, cf$,
$efdabc$,	$aebfcd$,
$fdeabc$,	$afcebd$,
$defbca$,	$adbecf$,
$efdbca$,	$aecdbf$,
$fdebca$,	$af \,.\, bd \,.\, ce$,
$defcab$,	$adcfbe$,
$efdcab$,	$ae \,.\, bf \,.\, cd$,

say

$$fdecab; \quad afbdce;$$

$$(ad \,.\, be \,.\, cf)\{(abc)\,\text{cyc.}\,(def)\,\text{cyc.}\} = (abcdef)_{18}.$$

See *post*, six letters, ord. 18.

The groups thus obtained, with substitutions which interchange the two sets of letters, are said to be “woven” groups.

By means of the foregoing notations a large number, but by no means all, of the groups belonging to the several cases of 2, 3, 4, 5, 6, 7, and 8 letters can be represented in a very compendious form. It is, even in the case of 4 letters, proper to introduce special notations: thus for the order 4, we have $(ab)(cd)$, (or as I generally write it $(ac)(bd)$), and $(abcd)$ cyc. (which it is convenient thus to represent), and another group $(1,\ ab\,.\,cd,\ ac\,.\,bd,\ ad\,.\,bc)$: this is a symmetrical character of the woven group $(ac\,.\,bd)(ab)(cd)$, but in thus representing it, we fail to exhibit the symmetrical character of the group, and I prefer to represent it as $(abcd)_4$. Again for the order 8, there is a single group, which I write $(abcd)_8$: this can be, in regard to dimidiation, divided in three different ways into two sets of four, and to distinguish them I write

$$(abcd)_8 \text{ com.} = (1,\ ac,\ bd,\ ac\,.\,bd;\ abcd,\ adbc,\ ab\,.\,cd,\ ad\,.\,bc),$$
$$(abcd)_8 \text{ cyc.} = (1,\ abcd,\ ac\,.\,bd,\ adbc;\ ac,\ bd,\ ab\,.\,cd,\ ad\,.\,bc),$$
$$(abcd)_8 \text{ pos.} = (1,\ ab\,.\,cd,\ ac\,.\,bd,\ ad\,.\,bc;\ ac,\ bd,\ abcd,\ adcb);$$

viz. com. denotes that the first set of four is the composite group $(ac)(bd)$; cyc. that it is the cyclical group $(abcd)$ cyc.; and pos. that it is the positive group $(abcd)_4$. We have thus, in the case of 6 letters, the three dimidiation forms,

$$\{(abcd)_8 \text{ com.}\,(ef)\} \text{ dim.} = (ac)(bd) + ef(abcd,\ adbc,\ ab\,.\,cd,\ ad\,.\,bc),$$
$$\{(abcd)_8 \text{ cyc.}\,(ef)\} \text{ dim.} = (abcd) \text{ cyc.} + ef(ac,\ bd,\ ab\,.\,cd,\ ad\,.\,bc),$$
$$\{(abcd)_8 \text{ pos.}\,(ef)\} \text{ dim.} = (abcd)_4 + ef(ac,\ bd,\ abcd,\ adcb),$$

the last of which may be more simply written as $\{(abcd)_8(ef)\}$ pos.

In the cases of 2, 3, 4, and 5 letters, the groups are as follows:

2 letters		3 letters		4 letters		5 letters	
ord.		ord.		ord.		ord.	
2	(ab)	3	(abc) cyc.	2	$(ab\,.\,cd)$	5	$(abcde)$ cyc.
		6	(abc) all	4	$(ab)(cd)$	6	(abc) cyc. (de)
				4	$(abcd)$ cyc.	10	$\{(abcd)_8(de)\}$ pos.
				4	$(abcd)_4$	10	$(abcde)_{10}$
				8	$(abcd)_8$	12	(abc) all (de)
				12	$(abcd)$ pos.	20	$(abcde)_{20}$
				24	$(abcd)$ all	60	$(abcde)$ pos.
						120	$(abcde)$ all

The nature of the group is in most cases at once intelligible from the foregoing explanations: thus (ab) denotes the group $(1,\ ab)$; (abc) cyc., the cyclical group $(1, abc, acb)$; (abc) all, the group of the six substitutions $(1, abc, acb, ab, ac, bc)$, and so in other cases: but, when a further explanation is required, the nature of the group is merely indicated by writing down the order as a suffix: thus $(abcd)_4$ means a group of four substitutions, $(abcd)_8$ a group of 8 substitutions, and so in other cases. In the case of four letters, the necessary explanations have already been given: it may be remarked that the group $(abcd)_8$ is that of the substitutions which leave unaltered the three-valued function $ac+bd$.

Five letters. Explanations. *Five letters.*

ord. 10. $(abcde)_{10}$. This may be written $\{(abcde)_{20}\}$ pos., viz. it consists of the 10 positive substitutions out of the next-mentioned group of 20 substitutions. Referring to that group, and writing $\sigma = S^2 = ac\,.\,bd$, and $T = aecdb$, the present group is $(1,\,\sigma)(1,\,T,\,T^2,\,T^3,\,T^4)$, where $\sigma^2 = 1$, $T^5 = 1$, $\sigma T = T^4\sigma$, $\sigma T^4 = T\sigma$, $\sigma T^3 = T^2\sigma$, $\sigma T^2 = T^3\sigma$: or, retaining S^2 instead of σ, say the group is $(1,\,S^2)(1,\,T,\,T^2,\,T^3,\,T^4)$.

ord. 20. $(abcde)_{20}$. The substitutions, distinguishing the positive and the negative ones, are

$$\begin{array}{llll}
+ & + & + & - \\
1, & ab\,.\,de, & abdce, & abcd, \\
 & ac\,.\,bd, & achbed, & abec, \\
 & ad\,.\,ce, & adebc, & acde, \\
 & ae\,.\,bc, & aecdb, & aceb, \\
 & bc\,.\,cd, & & adcb, \\
 & & & aebd, \\
 & & & aedc, \\
 & & & bced, \\
 & & & bdec;
\end{array}$$

and here, if $S = abcd$, $T = aecdb$, then the group is $(1,\,S,\,S^2,\,S^3)(1,\,T,\,T^2,\,T^3,\,T^4)$, where $S^4 = 1$, $T^5 = 1$, and generally $S^iT^j = T^k \cdot S^l$; that is,

	1	T	T^2	T^3	T^4
1	1	T	T^2	T^3	T^4
S	S	T^2S	T^4S	TS	T^3S
S^2	S^2	T^4S^2	T^3S^2	T^2S^2	TS^2
S^3	S^3	T^3S^3	TS^3	T^4S^3	T^2S^3

read

$$ST = T^2S, \quad S^2T^2 = T^3S^2, \text{ \&c.}$$

In the cases of 6, 7, and 8 letters, where the number of groups is larger, I introduce current numbers for the groups of the same order: thus, *infra*, 6 letters order 4, we have the groups 4. 1, 4. 2, 4. 3, 4. 4; and so in other cases, where there is more than one group for the same order.

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Six letters. Six letters: the groups are

ord.

2\. $(ab\,.\,cd\,.\,ef)$,

3\. $(abc\,.\,def)$ cyc.

4\. 1 $(ab\,.\,cd)(ef)$,

2 $\{(ab)(cd)(ef)\}$ pos.

3 $\{(abcd)$ cyc. $(ef)\}$ pos.

4 $\{(abcd)_4(ef)\}$ dim.

6\. 1 $(abcdef)$ cyc.

2 $(abc\,.\,def)$ all,

3 $(ad\,.\,bf\,.\,ce)(abc\,.\,def)$ cyc.

8\. 1 $(ab)(cd)(ef)$,

2 $(abcd)$ cyc. (ef),

3 $(abcd)_4(ef)$,

4 $\{(abcd)_8$ com. $(ef)\}$ dim.

5 $\{(abcd)_8$ cyc. $(ef)\}$ dim.

6 $\{(abcd)_8$ pos. $(ef)\}$ dim. $= \{(abcd)_8(ef)\}$ pos.

9\. (abc) cyc. (def) cyc.

12\. $(abcdef)_{12}$,

16\. $(abcd)_8(ef)$,

18\. 1 (abc) all (def) cyc.,

2 $\{(abc)$ all (def) all$\}$ pos.,

3 $(ad\,.\,be\,.\,cf)\{(abc)$ cyc. (def) cyc.$\}$,

24\. 1 $(abcd)$ pos. (ef),

2 $\{(abcd)$ all $(ef)\}$ pos.,

3 $(\pm\, abcdef)_{24}$ The $\pm$ and $+$ are used for distinction, the $\pm$ showing that the group is one with positive and negative substitutions, the $+$ that it is a group with positive substitutions only.

4 $(+\,abcdef)_{24}$,

36. 1 (abc) all (def) all,

2 $(abcdef)_{36}$,

48. 1 $(abcd)$ all (ef),

2 $(abcdef)_{48}$,

60. $(abcdef)_{60}$,

72. $(abcdef)_{72}$,

120. $(abcdef)_{120}$,

360. $(abcdef)$ pos.,

720. $(abcdef)$ all.

ord. 6. 2. $(abc\,.\,def)$ all; the substitutions are those of (abc) all each, compounded with the corresponding substitution of (def) all; viz. they are

$$\begin{array}{lll} 1, & ab\,.\,de, & abc\,.\,def, \\ & ac\,.\,df, & acb\,.\,dfe, \\ & bc\,.\,ef. & \end{array}$$

ord. 6. 3. $(ad\,.\,bf\,.\,ce)(abc\,.\,def)$ cyc. This is, in fact, the group $(ad\,.\,be\,.\,cf)(abc\,.\,dfe)$ cyc. explained in the introductory paragraphs; only for convenience the letters e, f have been interchanged. Making this interchange, the substitutions of the group $(ad\,.\,bf\,.\,ce)(abc\,.\,def)$ cyc. are

$$\begin{array}{lll} 1, & abc\,.\,def, & ad\,.\,bf\,.\,ce, \\ & acb\,.\,dfe; & ae\,.\,bd\,.\,cf, \\ & & af\,.\,be\,.\,cd. \end{array}$$

It may be added that, writing $\beta = ad\,.\,bf\,.\,ce$ and $\theta = abc\,.\,def$, the group is $(1,\, \beta)(1,\, \theta,\, \theta^2)$, where $\beta^2 = 1$, $\theta^3 = 1$, and $\beta\theta = \theta^2\beta$, $\beta\theta^2 = \theta\beta$. If $U = adbecf$, then $U^2 = abc\,.\,def = \theta$, and the form is $(1,\, \beta)(1,\, U^2,\, U^4)$: see $(abcdef)_{12}$, *infra.*

ord. 8. 4, 5, 6; the nature of these groups is explained in the introductory paragraphs.

ord. 12. $(abcdef)_{12}$. The substitutions are

$$\begin{array}{llllll} 1, & ab\,.\,ef, & ad\,.\,bf\,.\,ce, & abc\,.\,def, & adbecf, & \\ & ac\,.\,de, & af\,.\,be\,.\,cd, & & acb\,.\,dfe, & afcebd, \\ & bc\,.\,df, & ae\,.\,bd\,.\,cf, & & & \\ & & ae\,.\,bf\,.\,cd, & & & \end{array}$$

writing $U = adbecf$ and $\beta = ad\,.\,bf\,.\,ce$, this is

$$(1,\, \beta)(1,\, U,\, U^2,\, U^3,\, U^4,\, U^5),$$

where $U^6 = 1$, $\beta^2 = 1$, $\beta U = U^5\beta$,

$$\beta(1,\, U,\, U^2,\, U^3,\, U^4,\, U^5) = (1,\, U^5,\, U^4,\, U^3,\, U^2,\, U)\beta.$$

ord. 18. $(ad\,.\,be\,.\,cf)\{(abc)\,\text{cyc.}\,(def)\,\text{cyc.}\}$; this has been explained in the introductory paragraphs. Arranging them in a more convenient order, the substitutions are

$$\begin{array}{lllll}
1, & abc, & abc\,.\,def, & ad\,.\,be\,.\,cf, & adcfbe, \\
 & acb, & abc\,.\,dfe, & ae\,.\,bf\,.\,cd, & adbecf, \\
def, & acb\,.\,def, & af\,.\,bd\,.\,ce, & & aebfcd, \\
dfe; & acb\,.\,dfe; & & & aecdbf, \\
 & & & & afbdce, \\
 & & & & afcebd.
\end{array}$$

Six letters. Ord. 24. 3. $(\pm\, abcdef)_{24}$: the substitutions are

$$\begin{array}{lllll}
+ & + & + & - & - \\
1, & ac\,.\,bd, & abe\,.\,cdf, & abcd, & ab\,.\,cd\,.\,ef, \\
 & ac\,.\,ef, & abf\,.\,cde, & adcb, & ac\,.\,be\,.\,df, \\
 & bd\,.\,ef, & ade\,.\,bfc, & aecf, & ac\,.\,bf\,.\,de, \\
 & & adf\,.\,bec, & afce, & ad\,.\,be\,.\,cf, \\
 & & aeb\,.\,cfd, & bcdf, & ae\,.\,bd\,.\,cf, \\
 & & aed\,.\,bcf, & bfde, & af\,.\,bd\,.\,ce, \\
 & & afb\,.\,ced, & & \\
 & & afd\,.\,bce, & &
\end{array}$$

This includes, as part of itself, the group

$$\{(abcd)_8 \text{ cyc. } (ef)\} \text{ dim.};$$

and it may be written

$$(ade\,.\,bcf) \text{ cyc. } \{(abcd)_8 \text{ cyc. } (ef)\} \text{ dim.}$$

24. 4. $(+\, abcdef)_{24}$; the substitutions are

$$\begin{array}{llll}
1, & ab\,.\,cd, & abe\,.\,cdf, & abcd\,.\,ef, \\
 & ac\,.\,bd, & abf\,.\,cde, & adcb\,.\,ef, \\
 & ac\,.\,ef, & ade\,.\,bfc, & aecf\,.\,bd, \\
 & ad\,.\,bc, & adf\,.\,bec, & afce\,.\,bd, \\
 & ae\,.\,cf, & aeb\,.\,cfd, & bcdf\,.\,ac, \\
 & af\,.\,ce, & aed\,.\,bcf, & bfde\,.\,ac, \\
 & bd\,.\,ef, & afb\,.\,ced, & \\
 & be\,.\,df, & afd\,.\,bce, & \\
 & bf\,.\,de. & &
\end{array}$$

This includes, as part of itself, the group $\{(abcd)_8(ef)\}$ pos.; and it may be written $(ade\,.\,bfc)$ cyc. $\{(abcd)_8(ef)\}$ pos. It consists of the positive terms of $(abcdef)_{48}$, and might thus be written $(abcdef)_{48}$ pos.

ord. 36. $(abcdef)_{36}$; this consists of the positive terms of $(abcdef)_{72}$. The group consists of the substitutions which leave unaltered $b-c\,.\,c-a\,.\,a-b\,.\,e-f\,.\,f-d\,.\,d-e$.

ord. 48. $(abcdef)_{48}$. The substitutions are *Six letters.*

$+$	$+$	$+$	$+$	$-$	$-$
1,	$ab\,.\,cd$,	$abe\,.\,cdf$,	$abcd\,.\,ef$,	ac,	$abcd$,
	$ac\,.\,bd$,	$abf\,.\,cde$,	$adcb\,.\,ef$,	bd,	$adcb$,
	$ac\,.\,ef$,	$ade\,.\,cbf$,	$aecf\,.\,bd$,	ef;	$aecf$,
	$ad\,.\,bc$,	$adf\,.\,cbe$,	$afce\,.\,bd$,		$afce$,
	$ae\,.\,cf$,	$aeb\,.\,cfd$,	$bcdf\,.\,ac$,		$bcdf$,
	$af\,.\,ce$,	$aed\,.\,cfb$,	$bfde\,.\,ac$;		$bfde$;
	$bd\,.\,ef$,	$afb\,.\,ced$,			$af\,.\,ce\,.\,bd$,
	$be\,.\,df$,	$afd\,.\,ceb$;			$afdceb$;
	$bf\,.\,de$.				

$-$	$-$
$ab\,.\,cd\,.\,ef$,	$abecfd$,
$ac\,.\,bd\,.\,ef$,	$abecdf$,
$ac\,.\,be\,.\,df$,	$abfcde$,
$ac\,.\,bf\,.\,de$,	$adecbf$,
$ad\,.\,be\,.\,cf$,	$adfcbe$,
$ae\,.\,cf\,.\,bd$,	$aedcfb$,
$af\,.\,ce\,.\,bd$;	$afbced$;

ord. 60. $(abcdef)_{60}$; this consists of the positive terms of $(abcdef)_{120}$.

ord. 72. $(abcdef)_{72}$. The substitutions are

+	+	+	+	+
1,	abc,	$abc\,.\,def$,	$ab\,.\,de$,	$ad\,.\,bccf$,
	acb,	$abc\,.\,dfe$,	$ab\,.\,df$,	$ad\,.\,bfce$,
	def,	$acb\,.\,def$,	$ab\,.\,ef$,	$ae\,.\,bdcf$,
	dfe;	$acb\,.\,dfe$;	$ac\,.\,de$,	$ae\,.\,bfcd$,
			$ac\,.\,df$,	$af\,.\,bdce$,
			$ae\,.\,cf$,	$af\,.\,becd$,
			$bc\,.\,de$,	$bd\,.\,aecf$,
			$bc\,.\,df$,	$bd\,.\,afce$,
			$bc\,.\,ef$;	$be\,.\,adcf$,
				$be\,.\,afcd$,
				$bf\,.\,adce$,
				$bf\,.\,aecd$,
				$cd\,.\,aebf$,
				$cd\,.\,afbe$,
				$ce\,.\,adbf$,
				$ce\,.\,afbd$,
				$cf\,.\,adbe$,
				$cf\,.\,aebd$.

−	−	−	−	−
ab,	$ab\,.\,def$,	$ad\,.\,be\,.\,cf$,	$adbecf$,	
ac,	$ab\,.\,dfe$,	$ad\,.\,bf\,.\,ce$,	$adbfce$,	
bc,	$ac\,.\,def$,	$ae\,.\,bd\,.\,cf$,	$adcebf$,	
de,	$ac\,.\,dfe$,	$ae\,.\,bf\,.\,cd$,	$adcfbe$,	
df,	$bc\,.\,def$,	$af\,.\,bd\,.\,ce$,	$aebdcf$,	
ef;	$bc\,.\,dfe$,	$af\,.\,be\,.\,cd$;	$aebfcd$,	
	$de\,.\,abc$,		$aecdbf$,	
	$de\,.\,acb$,		$aecfbd$,	
	$df\,.\,abc$,		$afbdce$,	
	$df\,.\,acb$,		$afbecd$,	
	$ef\,.\,abc$,		$afcdbe$,	
	$ef\,.\,acb$;		$afcebd$;	

This contains, as part of itself, the group (abc) all (def) all; and it may be written $(ad\,.\,be\,.\,cf)\{(abc)\text{ all }(def)\text{ all}\}$. It leaves unaltered the function $abc + def$.

Six letters. ord. 120. $(abcdef)_{120}$. The substitutions are

+	+	+	+	−	−	−
1,	ab . cf,	abcef,	abc . dfe,	abcd,	abcfde,	ab . cd . ef,
	ab . de,	abdce,	abd . cfe,	abdf,	abdefc,	ab . ce . df,
	ac . bd,	abefd,	abe . cdf,	abec,	abedcf,	ac . be . df,
	ac . ef,	abfdc,	abf . cde,	abfe,	abfced,	ac . bf . de,
	ad . bf,	acbed,	acb . def,	acbf,	acbdfe,	ad . bc . ef,
	ad . ce,	acdfb,	acd . bfe,	acde,	acdbef,	ad . be . cf,
	ae . bc,	acedf,	ace . bfd,	aceb,	acefbd,	ae . bd . cf,
	ae . df,	acfbe,	acf . bde,	acfd,	acfedb,	ae . bf . cd,
	af . be,	adbcf,	adb . cef,	adbe,	adbfec,	af . bc . de,
	af . cd,	adbec,	adc . bef,	adcb,	adcebf,	af . bd . ce;
	bc . df,	adcfe,	ade . bfc,	adef,	adecfb,	
	bd . ef,	adfeb,	adf . bec,	adfc,	adfbce,	
	bc . cd,	aebdf,	aeb . cfd,	aebd,	aebcdf,	
	bf . ce,	aebfc,	aec . bdf,	aecd,	aecbfd,	
	cf . de;	aefcd,	aed . bcf,	aecf,	aedfcb,	
		aecdb,	aef . bdc,	aefb,	aefdbc,	
		afbde,	afb . ced,	afbc,	afbecd,	
		afcdb,	afc . bed,	afce,	afcdeb,	
		afdec,	afd . bce,	afdb,	afdcbe,	
		afecb,	afe . bcd;	afed,	afebdc;	
		bcdef,		bcfd,		
		bdfce,		bced,		
		becfd,		bdce,		
		bfedc;		bdfe,		
				becf,		
				bfcd,		
				bfde,		
				cdfe,		
				cefd.		